

C1 · Parsing-Retrieval Disconnect

Candidate parsers



Qwen3-VL



MinerU



Marker



PaddleOCR

Cleaner-looking ≠ better retrieval

parser	Boundary Clarity	answer	retrieval
MinerU	0.72 (highest)	no	MISS
Prod (ours)	lower	yes	HIT

highest Boundary Clarity, yet worst retrieval:
appearance metrics measure formatting, not content

anti-correlation $r = -0.81$

C2 / C3 · Diagnose + Select

C2 · Coverage diagnostic (no retriever)

where does each gold answer land?



covered

absent

absent = answer never made it into the parser output
(parser fault)
constant across all 8 chunkers → chunking can't recover it

C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3



mE5



Qwen3-Emb

→ rank parsers & chunkers by retrieval

C4 · Bounded Parser Training

C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

RADP-Distill — edit-distance → GT

RADP-DPO — page-local RCPS

Result

Hit@5 gain over the untuned parser:

RADP-Distill



+1.22 pp

RADP-DPO



+0.85 pp

≈ **overlapping CIs → equal gain**
retrieval reward unnecessary; lever = fidelity distillation